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FLEXURE AND TWIST TEST RELIABILITY ASSURANCE OF FLEXIBLE ELECTRONICS

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ABSTRACT

Additively Printed Flexible electronics in wearable application may be subjected to twisting or flexing depending on the form factor and the location of use. There is a dearth of standards for testing and reliability assurance of flexible electronics and reliability data for various use conditions. In this paper, a test protocol has been developed for twisting and flexing on aerosol-jet additively printed flexible circuits. Test patterns with commonly used traces geometries have been developed, aerosol-jet printed, and thermally sintered at various conditions. Reliability data has been acquired under both twist and flex using resistance monitoring. Failure mechanisms have been studied for both deformation modes.

Keywords: Polyimide substrate, Additive printing, Resistance, Aerosol Jet Printing, Failure, Reliability, Flexing, Twisting.

NOMENCLATURE

Ag	Silver
AJP	Aerosol-jet Printing
H1-H2	Trace 1 - 2 with trace geometry: Horseshoe
N	Number of cycles
R	Measured resistance, mOhm
R_0	Initial resistance, mOhm
S1-S2	Trace 1 - 2 with trace geometry: Straight
t	Thickness, mm
Z1-Z2	Trace 1 - 2 with trace geometry: Zigzag

1. INTRODUCTION

The trend towards weight reduction and miniaturization of electronics is driving the emergence of flexible electronics. A number of additive printed electronics methods are available including aerosol-jet printing, inkjet printing, screen printing, and gravure printing. Reduction in size and weight are especially important for applications focusing on sports, leisure, healthcare, military and security apparel, fashion, and wearable consumer electronics. [1]. Mechanical stresses are induced on flexible electronics when subjected to bending, which can lead to delamination, cracking or shearing of interconnects and additively printed layers [2]. An understanding of the shear strength of the deposited layer is important in addition to the effect of process conditions on the reliability and survivability of the printed structures. Aerosol-jet printing (AJP) is a popular method in flexible electronics because of its high-resolution in comparison with inkjet, screen printing. It has also gained much popularity as a custom low volume print electronics manufacturing technique. The vulnerability of the conductors for mechanical stresses needs to be investigated thoroughly as they are an essential block in flexible electronics [3]. From a bendability point of view, high mechanical strength and stability are key parameters required from the substrate material. A common choice for the substrate is plastic film, such as polyethylene terephthalate (PET), polyethylene naphthalate (PEN) or polyimide (PI) [4,5]. The relationship between microstructural evolution such as particle growth, and porosity and electrical conductivity during post heat treatment has been widely studied [4, 5]. Reducing the porosity and forming a dense microstructure can improve the electrical conductivity which can be attained by controlling annealing temperature [6-8], annealing time [8, 9], heat treatment method [10], and ambient atmosphere [11]. According to the end application, the process parameters can be varied in order to get a lower resistance but at the same time enough shear strength to suffice the need. Bending is one of the common body joint motion. Wearable application in flexible electronics being the major area it is important to analyze and improve reliability. In this paper, the reliability study is done of printed samples sintered at different conditions under cyclic bending and twisting motion. The conducting traces are printed with the help of AJP and then tested for its fatigue failure. The major reliability consideration is with respect to the wearable application. The resistances of the active traces are measured in-situ and recorded in real-time.

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2. MATERIALS AND METHODS

2.1 Test Vehicle

The test vehicle was designed at CAVE3 Electronics Research Center. On consideration of general geometries on PCB the following geometries viz. straight, horseshoe and zigzag patterns on the Flex-PCBs was finalized. The dimensions of the flexible test boards are 132mm x 76mm for the bending and 200mm x 76mm for the twisting condition. The Flex-PCB is made up of the polyimide substrate. The silver (Ag) traces are deposited on the substrate by additive printing using Aerosol Jet Printer (AJP) 300 system. Various geometries are been printed on Flex-PCB in order to analyze the effect of different geometry due to various test motions. Silver nanoparticle ink has been used for additive printing of the test patterns. The properties of the ink are shown in Table 1.

Ink Description:

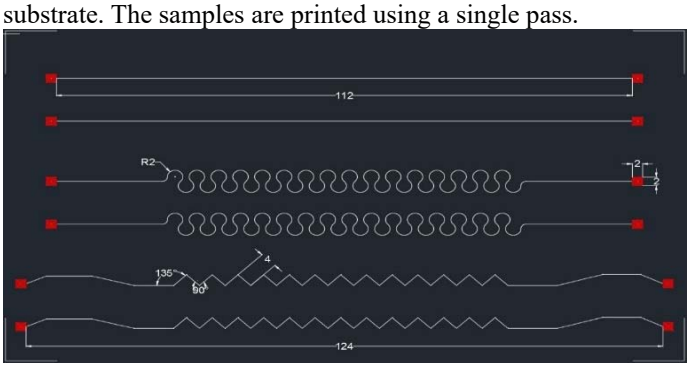
The particle size of silver ink is D (50) = 74nm; D (90) = 125nm. The viscosity of the ink is 33cP at 25°C. The cleaning process used for polyimide substrate is plasma cleaning for 5 minutes at 100W, O_2 clean.

Table 1: Ink composition

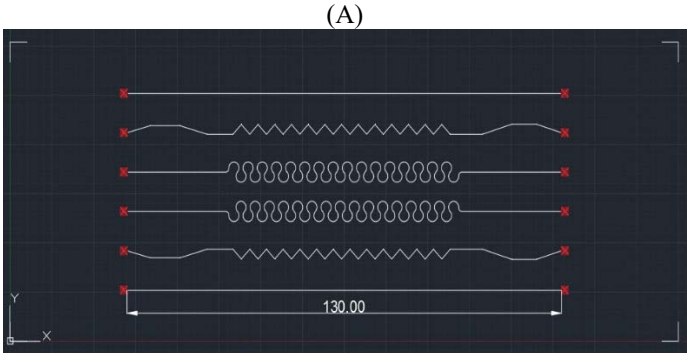
Product/Ingredient name	%
Silver	55-65
Propylene Glycol Monomethyl Ether	10-20
Propylene Glycol Monomethyl Ether Acetate	4-12
Diethylene Glycol Monomethyl Ether	5-10

Printing parameters:

The printing is done to achieve the line width of 0.15mm. The pneumatic atomizer is used for the deposition of ink on the substrate. The samples are printed using a single pass.



(A)



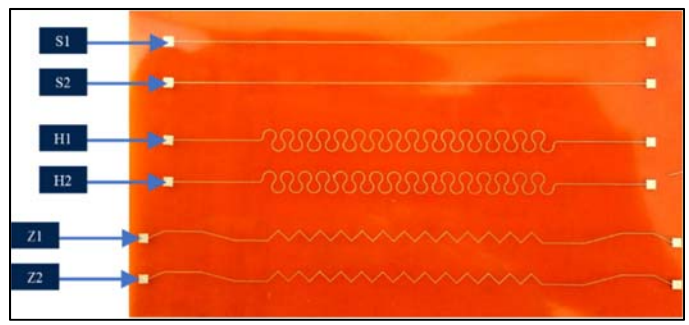
(B)

Figure 1: Trace Pattern (A) For Bending/Flexure (B) For Twisting

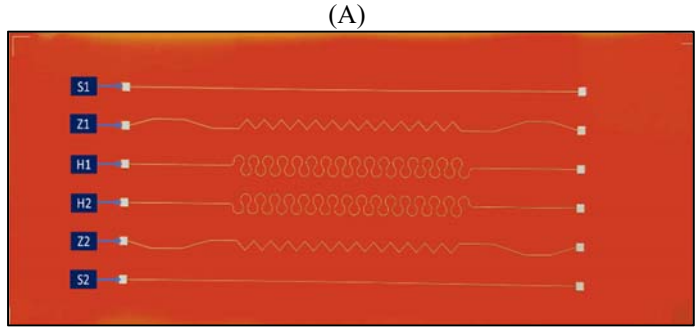
The trace patterns for twist and flexure are shown in Figure 1. The printing parameters used for fabrication of the specimen are shown in Table 2.

Table 2: Printing parameters

Process parameters	Value
PA MFC	800 ccm
Sheath Flow rate	40 ccm
Exhaust MFC	750 ccm
Speed	7 mm/s
Platen temperature	25°C
Ink temperature	25°C
Print Height	3mm
Nozzle diameter	300 µm



(A)



(B)

Figure 2: Nomenclature of The Test Vehicle (A) For Bending/Flexure (B) For Twisting

Each geometrically shaped trace has two identical traces in order to measure the consistency within the same sample. To study the effect of sintering temperature on the electrical and mechanical properties, after the printing is done the samples are sintered at 4 different temperatures. The samples are sintered for 1 hour at sintering temperatures 150°C, 200°C, 250°C and 300°C. Figure 3, shows the schematic of the pneumatic Aerosol printing. The pressure of the N_2 gas, mass flow rate of ink is been established using a detailed study of the process parameters. The optimal set of process parameters has then been used for fabrication of the samples used for twisting and flexing experiments in this paper.

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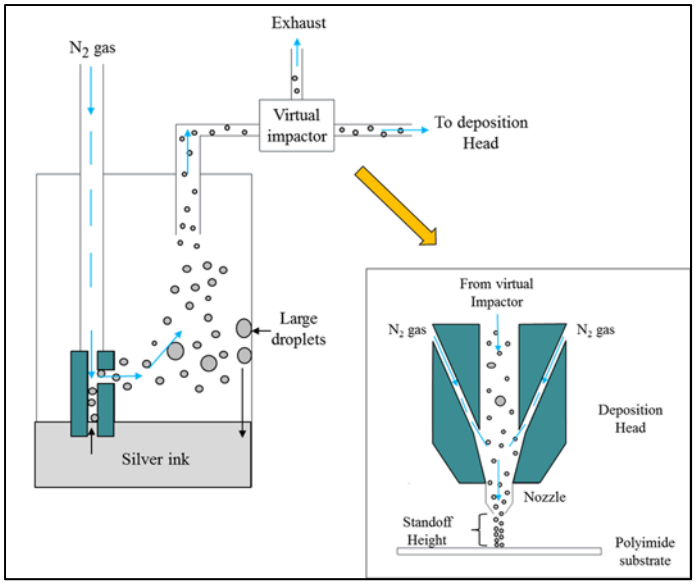

Figure 3: Pneumatic Atomizer Working

2.2 Experimental setup

Figure 4, shows the bending setup. The setup includes a Data acquisition unit connected to the test vehicle to measure the resistance across each individual trace which records the data into the computer. A stepper motor, power supply source, stepper motor driver, microcontroller, lead screw actuator and hinge setup are the parts in the experimental setup. The acquisition unit measures real-time data and records it into the software, which helps in analyzing the fatigue failure point more conveniently.

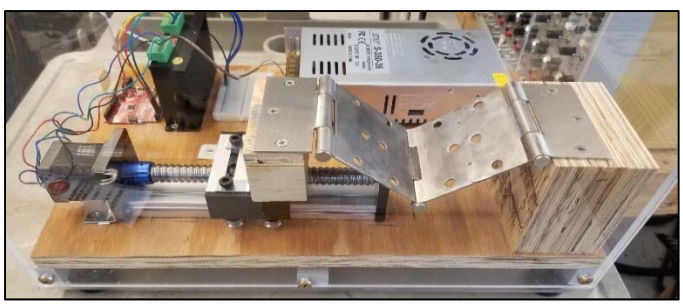

Figure 4: Experimental Setup (Bending/Flexure)

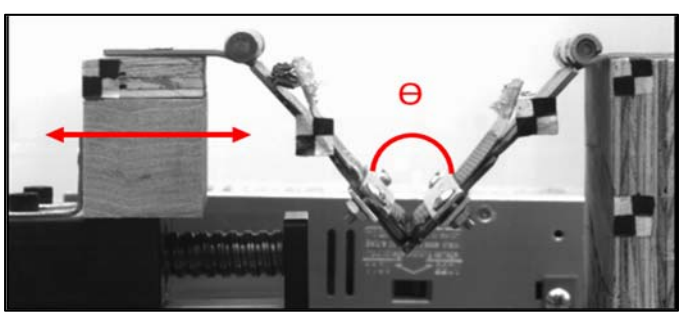

Figure 5: Motion Of The Test Board (Bending/Flexure)

Figure 5 and Figure 7, explains the motion of setup. The test setup is being tested for a v- bend considering the body motions like folding hands at elbow or folding of legs and twist motions like wrist rotation. These bends and twists have stress at a region or over a region of line and not on the entire test board. (Lall 2018). The wires are connected to the traces on the substrate using silver conductive epoxy. Figure 5 shows how the actuator moves in the horizontal direction giving the hinge a circular motion. The samples are fixed with the help of 2 metal sheets of width 4mm at 10mm from the center of the hinge.

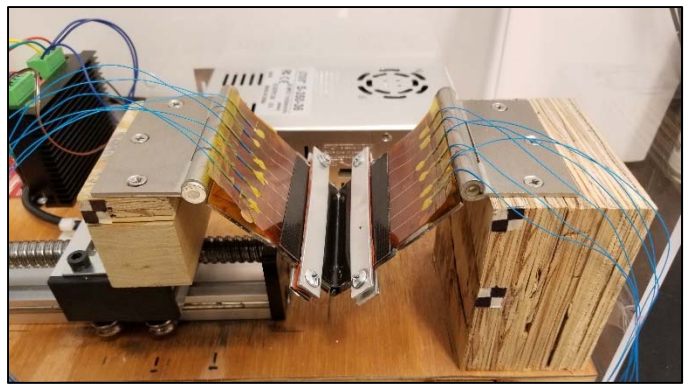

Figure 6: Experimental Setup With A Test Vehicle (Bending/Flexure)

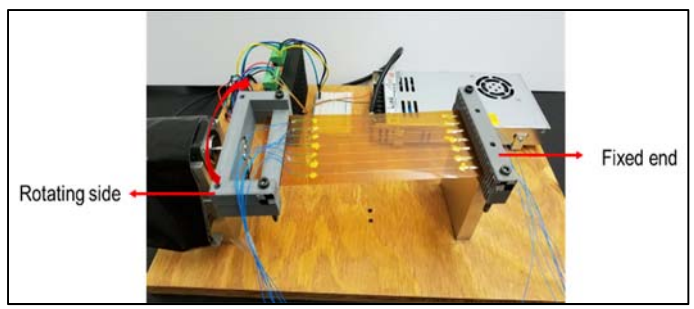

Figure 7: Experimental Setup With A Test Vehicle (Twisting)

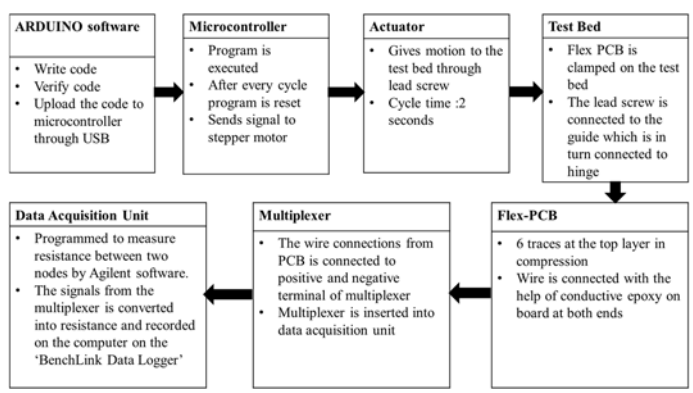

Figure 8: Block Diagram Of The Experimental Setup

Figure 8, shows the experimental stepwise flow. The two ends of the traces are connected wire using silver conductive epoxy. The wire connection is routed to the multiplexer (34901A) which is connected to the Agilent Data Acquisition Unit (34970A). The acquisition unit is connected to a computer through GPIB-USB

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cable. Cycle time for the setup is 2 secs (0.5 Hz). The acquisition unit takes readings at an interval of 1sec such that it takes reading at the start of the cycle and at the bent position similarly in twist state. The data recorded by the software is saved with '.csv', an extension which can be imported to the excel for analysis. Figure 9, shows the actual graph plot using the high-speed camera for its position vs time graph to measure frequency and velocity.

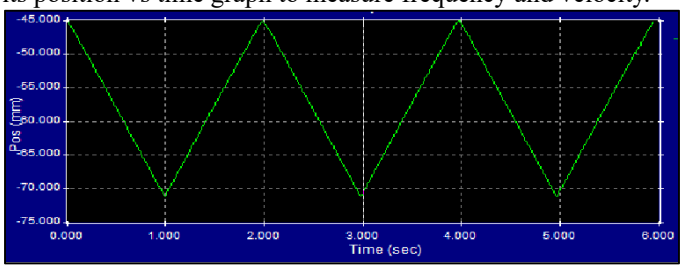

Figure 9: Graph Plot For Frequency

Test Matrix

For Flexural test:

Substrate dimension: 132mm x 76mm
Frequency: 0.5 Hz
Initial angle = 100˚
Final angle = 80˚

Table 3: Test Matrix For Flexural Test

Flexure Test				
ΔΘ = 20°	Sintering Temperature @ 1 hour			
	150C	200C	250C	300C

For Twist test:

Substrate dimension: 200mm x 76mm
Frequency: 0.5 Hz
Initial angle = +25˚
Final angle = -25˚

Table 4: Test Matrix For Twist Test

Twist Test				
Θ = +25° to -25°	Sintering Temperature @ 1 hour			
	150C	200C	250C	300C

3. RESULTS AND DISCUSSION

The flex-PCBs sintered at different temperatures are tested on the bending and twisting machine. The traces on the substrate undergo compression stress state during bending and torsional load during twisting. The samples are initially fixed on the test bed at 100˚. The bending cycle is from 100˚ to 80˚ (ΔΘ = 20˚). Whereas in case of twist the substrate is initially positioned at Θ. The failure criteria of the test board considered are 20% increase in resistance from its initial value. All samples are tested for 100,000 cycles for bending cycle. All the traces do not fail in this duration of time.

3.1 Effect of sintering temperature on resistance

Figure 10, shows the effect of varying sintering temperature on resistance keeping the sintering time same. It can be observed from the below graph that as the temperature of sintering is increased the resistance of the traces goes on decreasing. During the sintering, the nanoparticles of the ink try to come closer due to evaporation of solvents. In the samples tested the resistance value tends to flatten out after 250ºC regardless of the geometrical shape.

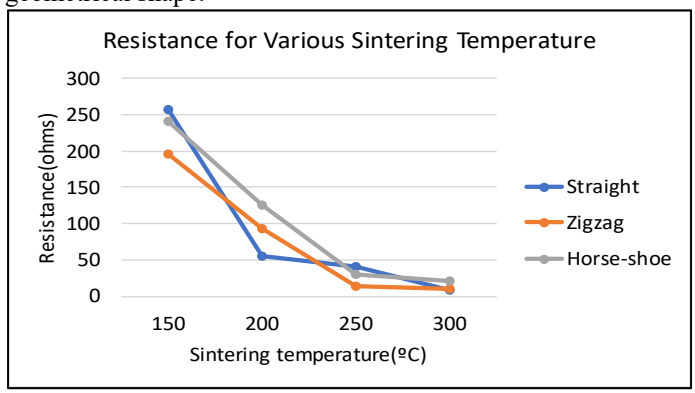

Figure 10: Effect Of Various Sintering Temperature On Resistance

3.2 Effect of sintering temperature on shear load to failure

Figure 11**,** shows the relation of sintering temperature and shear load to failure. As we can see that with an increase in temperature the shear load to failure tends to increase initially and then it starts reducing, similar data could be seen from the previous data published. The shear load to failure is highest at 250 ºC. The test is conducted with the help of a shear tester machine.

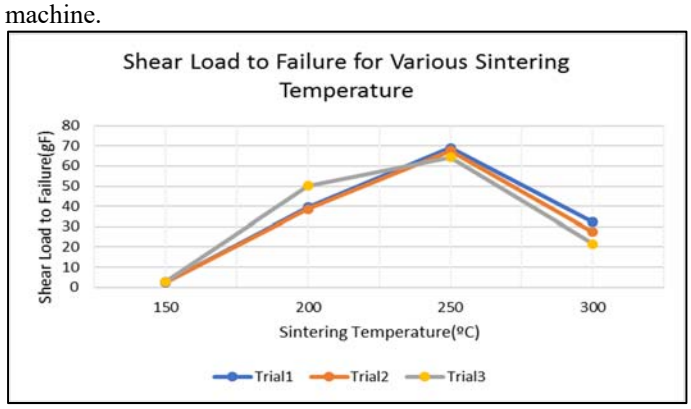

Figure 11: Effect Of Various Sintering Temperatures On Shear Load To Failure

3.3 Cyclic Bending Results

Each sample was tested for 100,000 cyclic bending. The change in resistance was recorded with the help of the data acquisition unit. For the test conditions it is observed that the resistance of the traces gradually increases over the number of cycles as per the work by Joon [2]. The resistance of the traces increases due to crack formation in the region of the bend. The fluctuation in the graphs shown below is due to the connection it establishes while bending and losing when it is at the initial condition. Continuous bending action leads to wrinkles on the flexible substrate there by accelerating the crack formation process. The resistance of the traces has an increasing trend with

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the number of cycles to failure. Figure 12- Figure 23 show the test results are for the testing condition of initial angle of 100º and final angle of 80º.

Samples sintered at 150 ˚C for 1 hour

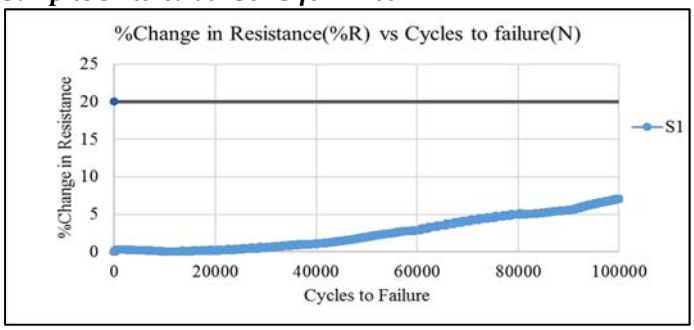

Figure 12: Data On Trace S1 For Cyclic Bending For One Of The Samples

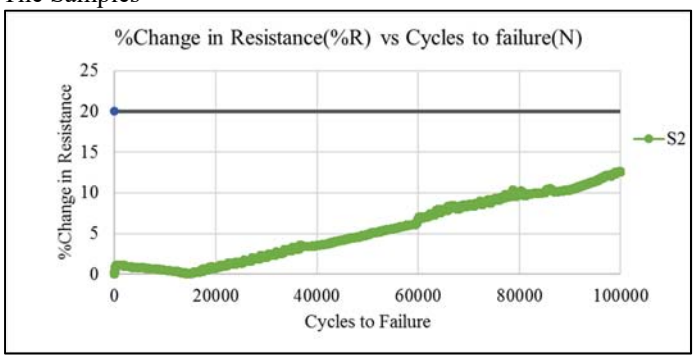

Figure 13: Data On Trace S2 For Cyclic Bending For One Of The Samples

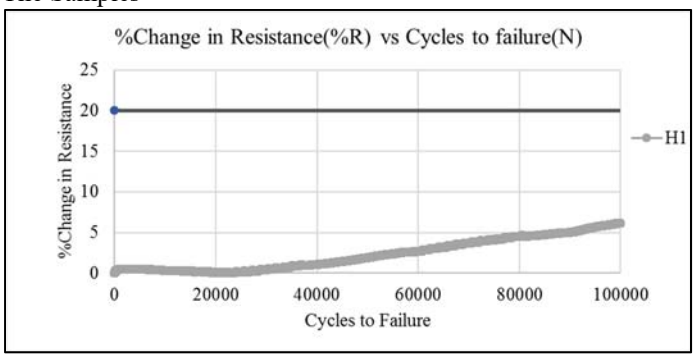

Figure 14: Data On Trace H1 For Cyclic Bending For One Of The Samples

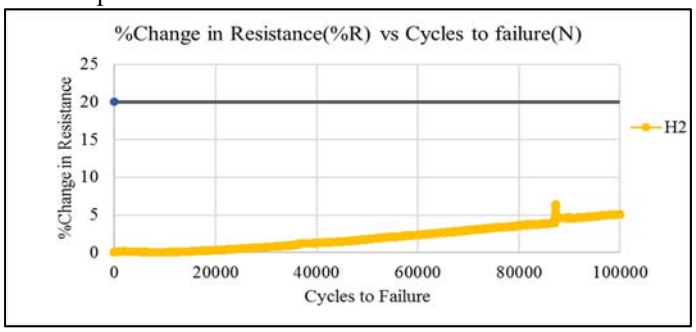

Figure 15: Data On Trace H2 For Cyclic Bending For One Of The Samples

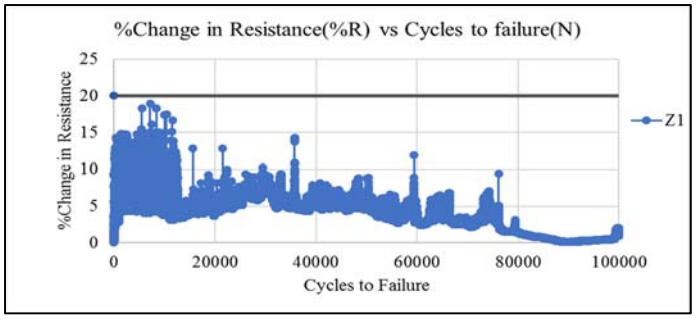

Figure 16: Data On Trace Z1 For Cyclic Bending For One Of The Samples

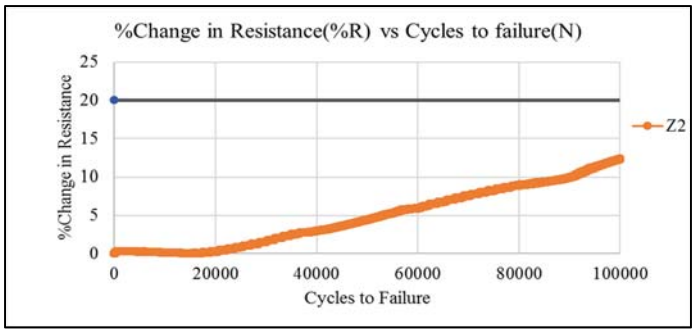

Figure 17: Data On Trace Z2 For Cyclic Bending For One Of The Samples

Samples sintered at 200 ˚C for 1 hour

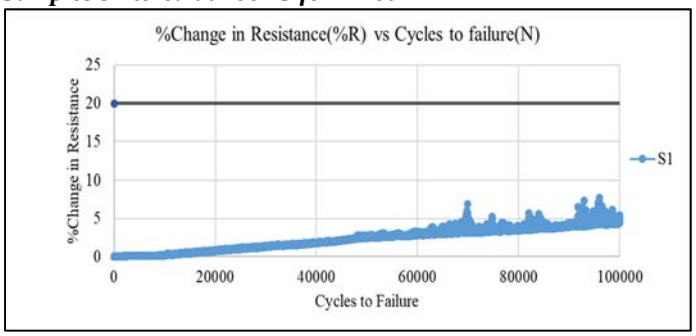

Figure 18: Data On Trace S1 For Cyclic Bending For One Of The Samples

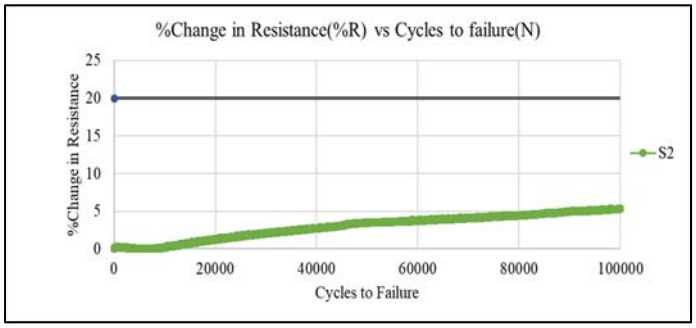

Figure 19: Data On Trace S2 For Cyclic Bending For One Of The Samples

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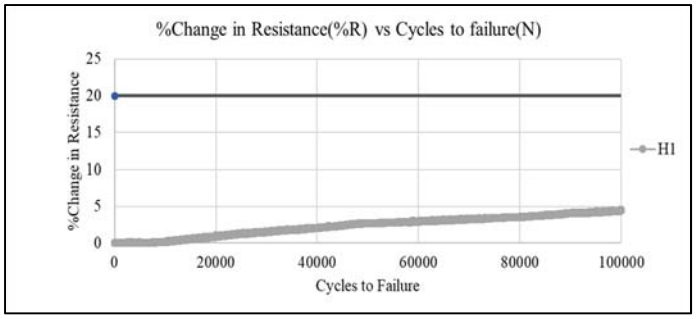

Figure 20: Data On Trace H1 For Cyclic Bending For One Of The Samples

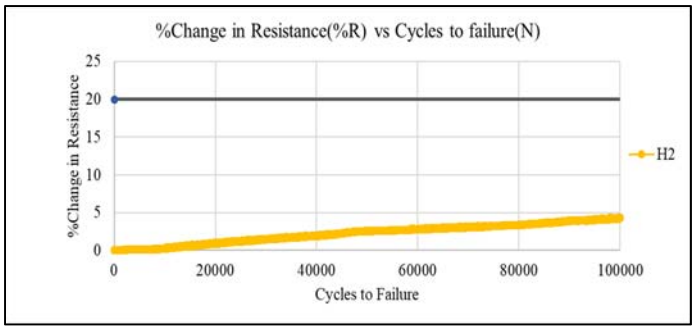

Figure 21: Data On Trace H2 For Cyclic Bending For One Of The Samples

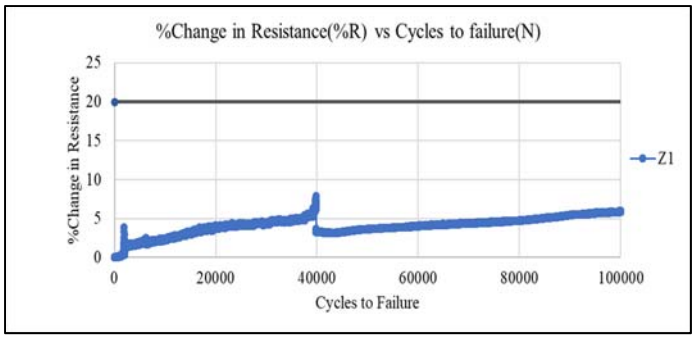

Figure 22: Data On Trace Z1 For Cyclic Bending For One Of The Samples

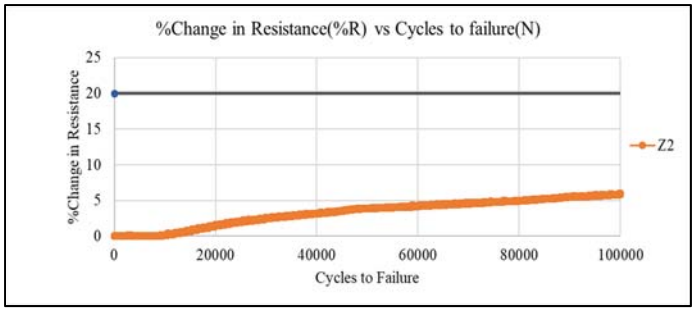

Figure 23: Data On Trace Z2 For Cyclic Bending For One Of The Samples

Comparing the results of samples sintered at 150 ˚C and 200 ˚C it can be noticed that the slope of the resistance graph goes on flattening with increasing sintering temperature in each geometry. The reason for this is the shear force required for failure in 200 ˚C sintered sample is much higher as compared to 150 ˚C, thus it can be related to the behavioral pattern of the shear load to failure data.

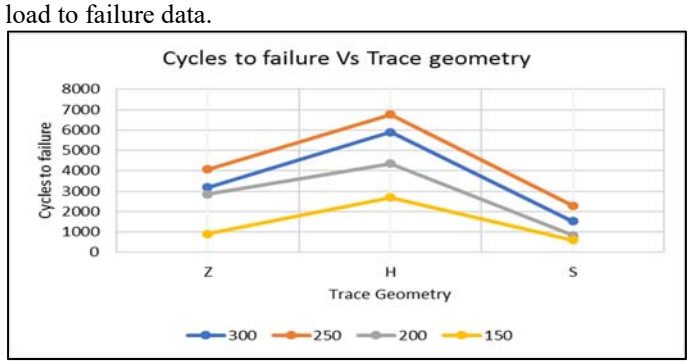

Figure 24: Comparison of Trace Geometries (Lall 2019)

Figure 24, show the data of the samples tested at initial angle of 100º and final angle of 60º. In this condition the samples tend to fail before 8000 cyclic bending whereas in the current test conditions it was observed that there are no samples failed in the 100,000 testing cycles. Thus, the reduction of angle variation by 20º improves the life of the tested sample. One more major observation which also relates to the prior study that the horseshoe bend geometry is less susceptible to failure as compared to rest of the geometries. Further test cycles need to be run to quantify the cycles to failure for bending condition.

3.4 Cyclic Twisting Results

Each sample was tested for 100,000 for cyclic twist motion as well. Similar kind of trend can be observed in this case as well of the resistance exhibits monotonic increase with the increase in twist cycles. The major difference observed in this motion pattern is the amount of stress on the substrate is less as compared to the bending motion, which leads to a relatively lower increase in resistance. The variation of the resistance change with respect to its twist motion is been shown below. The below graph is of one of the samples sintered at 250 ˚C for 1 hour.

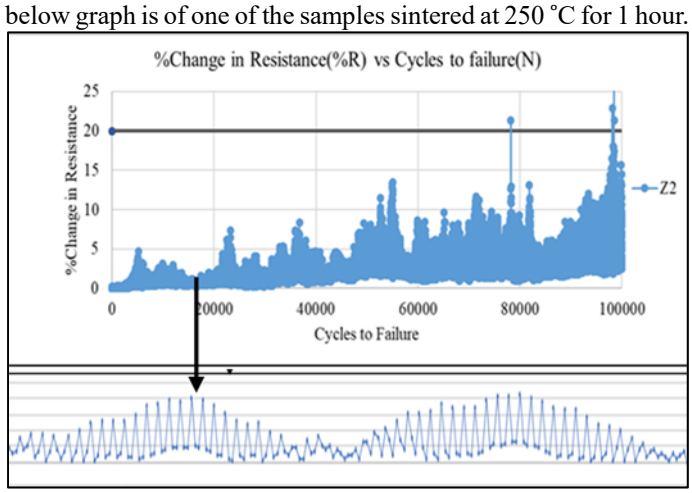

Figure 25: Data Of One Of The Samples From Cyclic Twisting With Resistance Variation Pattern During Testing

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Samples sintered at 250 ˚C for 1 hour

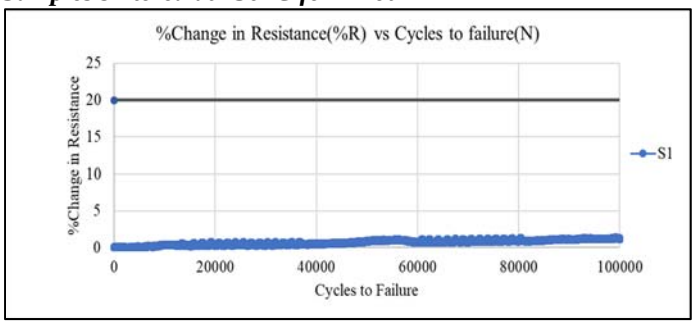

Figure 26: Data On Trace S1 For Cyclic Bending For One Of The Samples

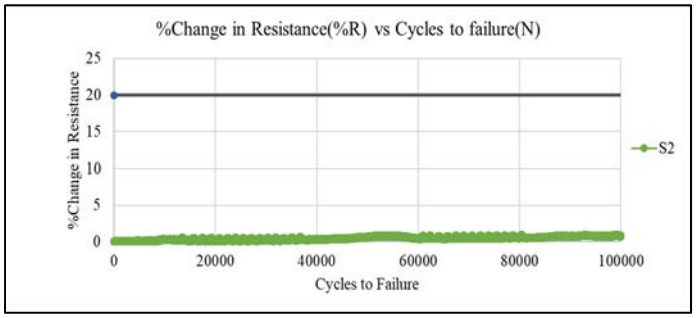

Figure 27: Data On Trace S2 For Cyclic Bending For One Of The Samples

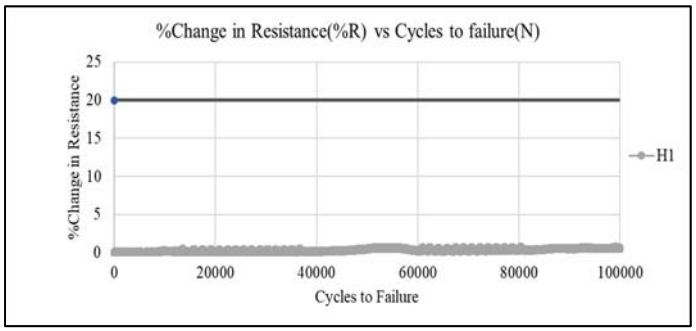

Figure 28: Data On Trace H1 For Cyclic Bending For One Of The Samples

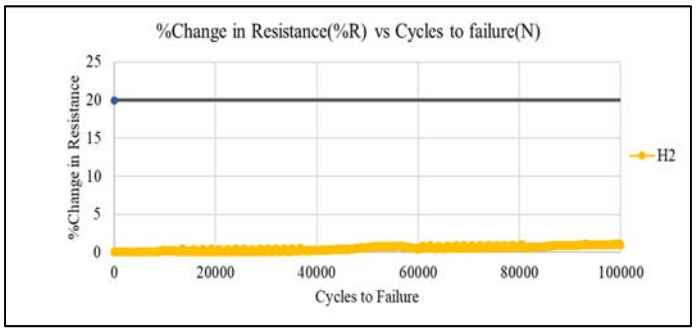

Figure 29: Data On Trace H2 For Cyclic Bending For One Of The Samples

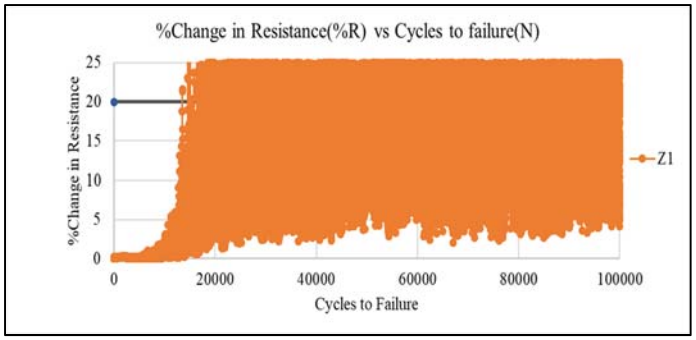

Figure 30: Data On Trace Z1 For Cyclic Bending For One Of The Samples

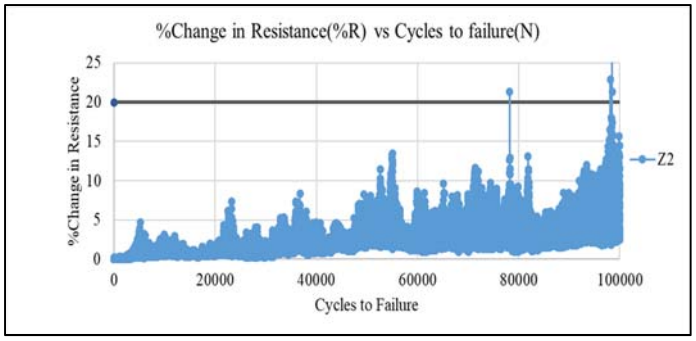

Figure 30: Data On Trace Z2 For Cyclic Bending For One Of The Samples

Samples sintered at 300 ˚C for 1 hour

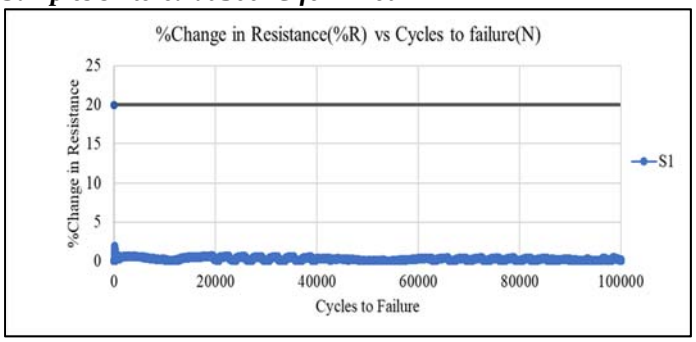

Figure 31: Data On Trace S1 For Cyclic Bending For One Of The Samples

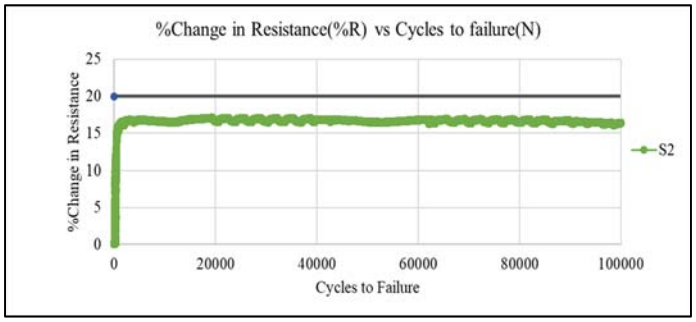

Figure 32: Data On Trace S2 For Cyclic Bending For One Of The Samples

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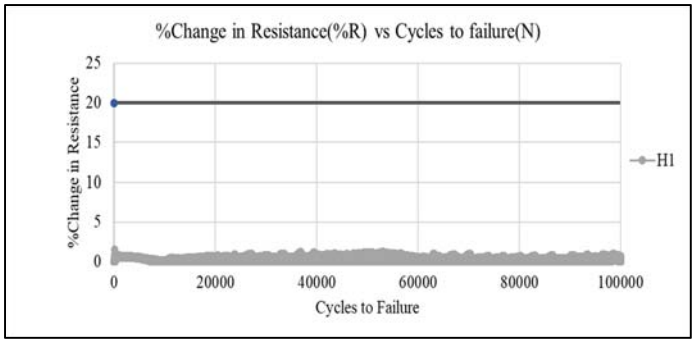

Figure 33: Data On Trace H1 For Cyclic Bending For One Of The Samples

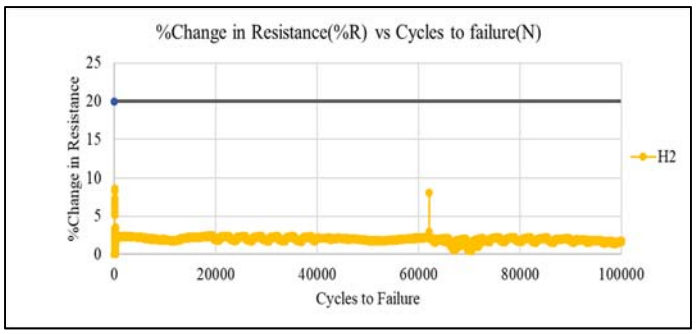

Figure 34: Data On Trace H2 For Cyclic Bending For One Of The Samples

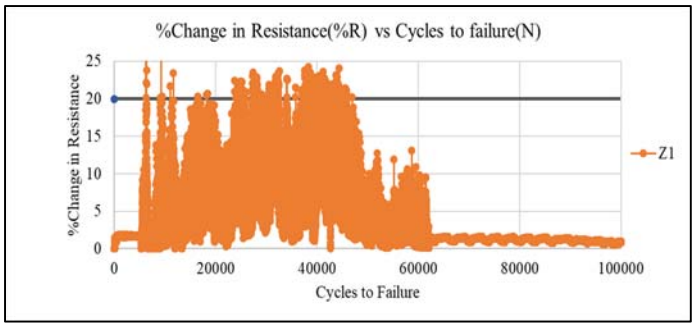

Figure 35: Data On Trace Z1 For Cyclic Bending For One Of The Samples

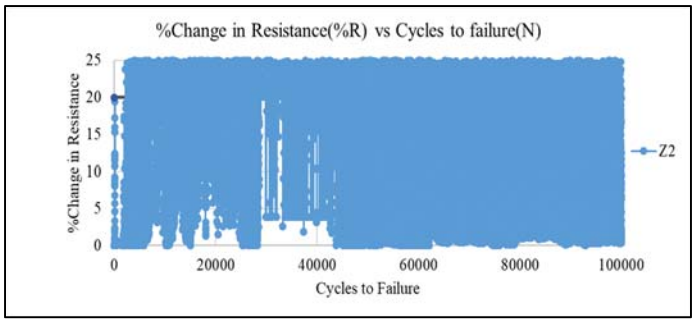

Figure 36: Data On Trace Z2 For Cyclic Bending For One Of The Samples

From both the sintering condition result it can be noticed that the traces with zigzag geometry failed but other two geometry sustain the torsion. The reason for this geometry to fail earlier than rest two is due to the direction of the force getting applied due to twisting. During twisting the zigzag pattern tends to experience a normal force twice in one cycle each at approximately 90˚ apart.

3.5 Optical images

From the below images it can be observed that the trace starts settling down with increasing temperature as the solvent goes on evaporating. This result compares with the study in the paper [9]. Similar results are observed in zigzag and horse-shoe geometrical traces.

Straight traces:

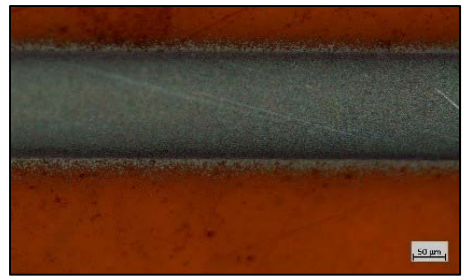

Figure 37: 150°C For 1 Hour

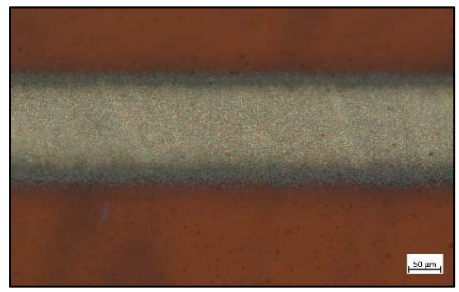

Figure 38: 200°C For 1 Hour

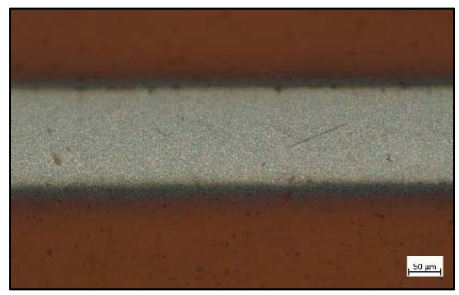

Figure 39: 250°C For 1 Hour

For the below images the filter has been changed to get a better image of the traces and its failure/defect areas.

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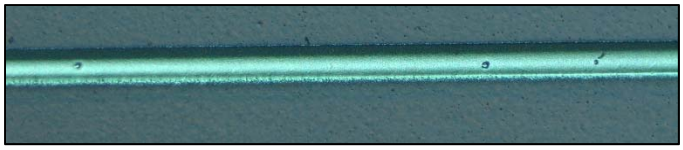


Figure 40: Pristine Image Of Straight Geometry Trace

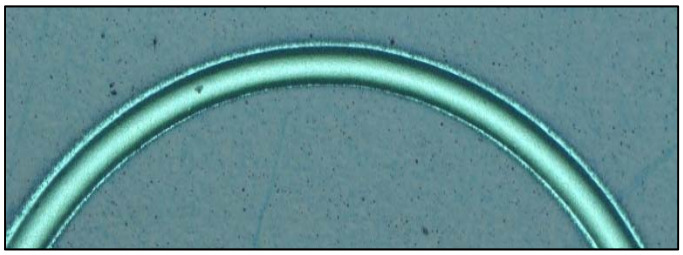


Figure 41: Pristine Optical Image Of Horseshoe Bend Geometry Trace

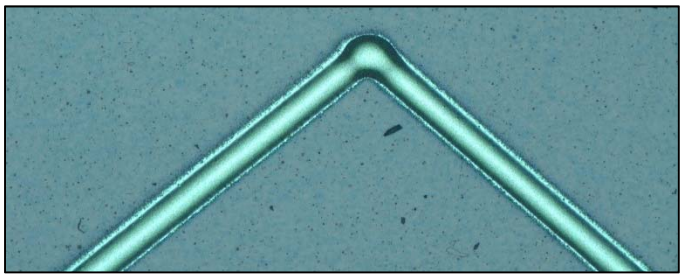


Figure 42: Pristine Optical Image Of Zigzag Geometry

3.6 Optical results of Traces after testing

For bending

The samples after cyclic bending undergo deformation and micro-cracks which leads to an increase in the resistance of the trace. From Figure 43-Figure 45, it can be seen that the failure is due to cracks.

For twisting

As seen from the below figures the failure of the samples due to twisting is different from the samples failed due to bending. In the twisting condition the failure of the sample is due to shear force from two directions. This phenomenon lead to delamination of interconnect from the substrate as compared to just crack formation as seen in the earlier test motions.

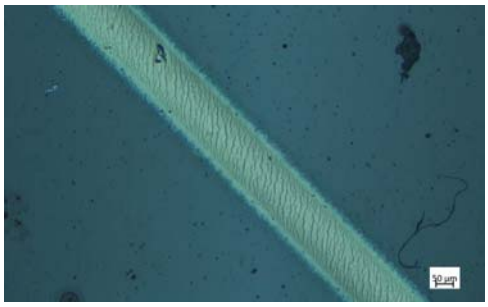


Figure 43: Straight Trace Of One Of The Samples After Failure

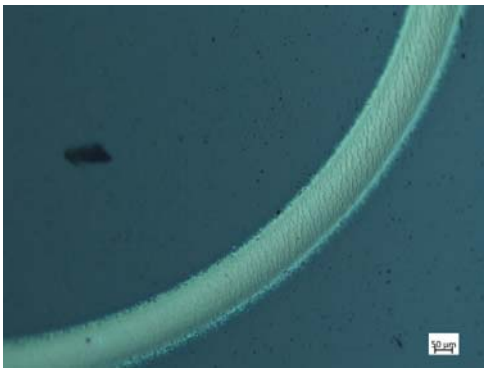


Figure 44: Horseshoe Trace Of One Of The Samples After Failure

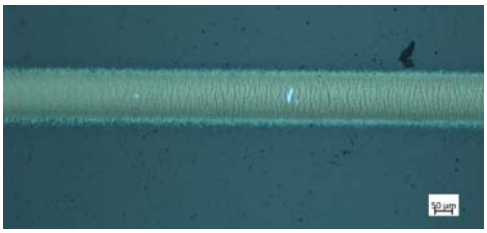
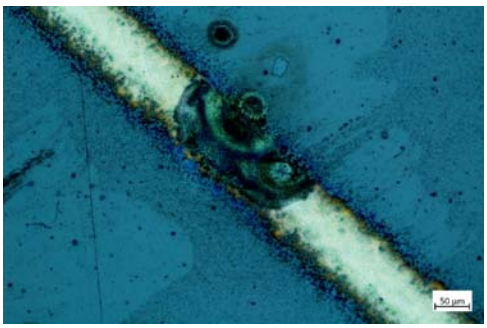
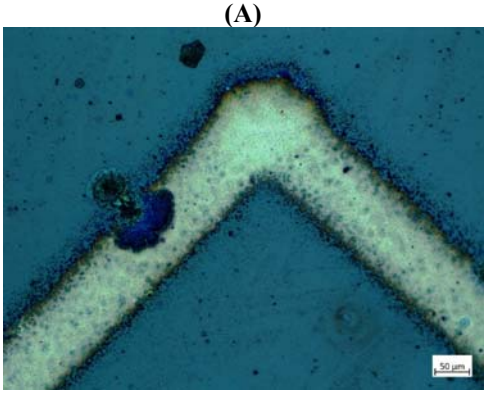


Figure 45: Zigzag Trace Of One Of The Samples After Failure



(A)



(B)

Figure 46: Zigzag Trace Of One Of The Samples After Failure Due To Both Side Twist In Part (A) And (B)

4. Summary and Conclusions

In this paper, two different types of motion including twist and flexing has been studied for the effect of sintering

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temperature on the printed interconnects. In addition, the effect of sintering on the shear strength of the interconnects has been quantified. Data of resistance and shear load indicates that conductive traces printed with using silver nanoparticle have a lower susceptibility to failure at 250˚C. The cyclic bending test results adds further evidence to the same. It can be observed that with reducing the final angle by 20˚ makes a big difference in the cycles to failure. The samples can sustain more than 100,00 cycles as compared to 8000 cycles in the prior study. The twisting motion too follows a similar trend and can be noticed that the amount of stress induced on the interconnects is much less as compared to the bending motion. It is also evident that the zigzag traces tend to fail early as compared to other geometry reason being the normal stress acting on the trace was higher as compared to other trace geometry. It was also observed that the polyimide substrate too undergoes plastic deformation at the end of 8000 cycles of flexure.

ACKNOWLEDGEMENTS

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